# Research Evolution — Project 1 (Tesla, TSLA)

**Erlan Jones** · **FIN 43900.** This file contains Edition A as submitted at the checkpoint (unchanged except the receipt identifier, recorded afterward as its preservation instruction requires), followed by the final Edition B starting on a new page.

# Project 1 — Edition A: Tesla, Inc. (TSLA)

**Decision user:** Investment Committee
**Valuation date:** September 1, 2026
**Initial thesis: Watch / defer.** Tesla has a substantial liquidity position and a faster-growing energy storage business, but vehicle deliveries, automotive margins, AI-related capital intensity, and commercialization risk make a near-term initiation premature. Initiation requires evidence that the energy and AI opportunity can offset automotive pressure without eroding returns on capital.

## Primary filing and known evidence

**Filing:** Tesla, Inc., Form 10-K for the fiscal year ended December 31, 2025; filed January 29, 2026; accession `0001628280-26-003952`. SEC filing index · Primary 10-K

| Item | Evidence | Filing source |
|---|---|---|
| Entity | Tesla, Inc. (NASDAQ: TSLA) | Cover page |
| Reporting period | Fiscal year ended December 31, 2025 | Cover page |
| Filing date | January 29, 2026 | SEC filing index |
| Currency and units | U.S. dollars; amounts in millions unless otherwise noted | 10-K financial statements |
| Diluted-share evidence | 3,528 million FY2025 weighted-average diluted shares; 303 million incremental shares from stock-based awards | Note 2—EPS |
| Cash and cash equivalents | $16,513 million at December 31, 2025 | Note 4—Cash and investments |
| Short-term investments | $27,546 million at December 31, 2025 | Note 4—Cash and investments |
| Debt | $8,180 million aggregate principal; $1,580 million current | MD&A—liquidity / Note 9—Debt |
| Operating-value | Energy generation and storage revenue grew 27% to | MD&A—energy |

| Item | Evidence | Filing source |
|---|---|---|
| driver | $12,270 million; segment gross margin was 29.8% | generation and storage |

The filing reports FY2025 revenue of $94.827 billion, down from $97.690 billion in FY2024. Tesla delivered approximately 1.64 million consumer vehicles and deployed 46.7 GWh of energy storage products in 2025.

## Synthetic enterprise-to-equity bridge check

All values below are U.S. dollars in millions. The class enterprise-value input and known answer were not provided, so the numerical check remains open rather than using an invented value. This bridge treats cash and short-term investments as non-operating financial assets and debt at aggregate principal; lease liabilities, noncontrolling interests, and other claims require the convention used in class.

| Bridge item | Amount | Evidence / calculation |
|---|---|---|
| Enterprise value | [Class input required] | Known class exercise input |
| Less: debt | (8,180) | FY2025 10-K, liquidity / Note 9 |
| Add: cash and cash equivalents | 16,513 | FY2025 10-K, Note 4 |
| Add: short-term investments | 27,546 | FY2025 10-K, Note 4 |
| Equity value | [EV – 8,180 + 16,513 + 27,546] | Calculation |
| Diluted shares | 3,528 | FY2025 10-K, Note 2 |
| Implied value per diluted share | [Equity value ÷ 3,528] | Calculation |
| Known-answer check | Pending class input | Reconcile after class EV is supplied |

## Consequential assumptions

- The filing’s energy-storage growth and margin expansion are durable despite tariffs, product mix, and deployment timing.
- Automotive delivery growth, manufacturing utilization, and price/mix can support automotive margins while Tesla funds new product and AI programs.
- AI-related investments, including autonomy, robotics, data centers, and fleet assets, produce attractive incremental returns; Tesla expects capital expenditures to exceed $20 billion in 2026.
- Cash and short-term investments are fully included in the bridge and aggregate debt principal is the applicable debt measure; the class convention may require adjustments.

## Unknowns and research plan

1. Obtain the class enterprise-value input and known answer; calculate and reconcile the bridge.
2. Track vehicle deliveries, automotive gross margin, average selling price, regulatory-credit revenue, and factory utilization in subsequent SEC filings.
3. Test energy-storage growth and gross margin against tariffs, unit economics, deployment volume, and capital needs.
4. Evaluate the timing, regulatory feasibility, and unit economics of autonomy, Robotaxi, and other AI-related initiatives.
5. Develop a valuation range using separate automotive and energy/AI scenarios, with explicit dilution and capital-expenditure assumptions.

## Falsification question

**Partner-generated question — answer intentionally omitted:** What evidence would show that Tesla's expected future growth and profitability are not strong enough to justify its current market valuation?

## Checkpoint receipt

**Submission file:** `Project-1-Edition-A.md`
**Receipt identifier:** Brightspace quiz "Project 1 — Edition A Checkpoint (AI Closed)", attempt 1 of 3, completed September 1, 2026 (window 1:30–6:00 PM).
**Preservation instruction:** After submission, retain this exact file unchanged and record the receipt identifier here.

# Project 1 — Edition B: Tesla, Inc. (TSLA) — Company Research Report

**Decision user:** Investment Committee
**Valuation date:** September 1, 2026
**Primary source:** Tesla, Inc. Form 10-K for the fiscal year ended December 31, 2025, filed January 29, 2026, accession `0001628280-26-003952`. SEC filing

## Business and investment context

Tesla operates automotive and energy generation and storage segments and is pursuing growth in autonomy, Robotaxi, and robotics. Management's strategy is to use its vehicle, energy, software, and AI capabilities to build a more service-oriented business. [Source: Tesla FY2025 Form 10-K, Item 1, p. 2; Item 7, p. 31]

The investment case has two opposing forces. The automotive business weakened in 2025, while the energy generation and storage business grew quickly and reported stronger segment economics. The central question is whether the newer growth initiatives can offset pressure in automotive without requiring uneconomic investment. [Source: Tesla FY2025 Form 10-K, Item 7, pp. 31, 37–38]

## Operating performance

FY2025 revenue was **$94.827 billion**, down 3% from $97.690 billion in FY2024. Automotive revenue declined 10% to $69.526 billion; automotive sales revenue declined 9%, which management attributed to lower cash deliveries and lower average selling prices driven by sales mix and customer incentives. [Source: Tesla FY2025 Form 10-K, Item 7, p. 37]

Tesla produced approximately **1.66 million** consumer vehicles and delivered approximately **1.64 million** in 2025. Net income attributable to common stockholders was **$3.794 billion**, down from $7.091 billion in 2024. [Source: Tesla FY2025 Form 10-K, Item 7, p. 31; Note 2, p. 61]

Energy generation and storage was the principal positive operating driver: segment revenue increased 27% to **$12.771 billion**, led by higher Megapack and Powerwall deployments. Segment gross margin increased to **29.8%** from 26.2%, and Tesla deployed **46.7 GWh** of energy-storage products in 2025. [Source: Tesla FY2025 Form 10-K, Item 7, pp. 31, 37–38]

## Financial position and capital needs

Tesla ended 2025 with **$16.513 billion** of cash and cash equivalents and **$27.546 billion** of short-term investments. Aggregate debt principal was **$8.180 billion**, including $1.580 billion due within one year. [Source: Tesla FY2025 Form 10-K, Item 7, p. 41; Note 4, pp. 62–63]

Operating cash flow was **$14.747 billion** in 2025, while capital expenditures were **$8.53 billion**. Management expects capital expenditures to exceed **$20 billion** in 2026, driven by AI compute and data centers,

manufacturing and R&D capacity, company-operated AI-enabled assets, and service and charging infrastructure. [Source: Tesla FY2025 Form 10-K, Item 7, pp. 31, 41–42]

Tesla reported **3.528 billion** weighted-average diluted shares in FY2025, including a **303 million** diluted impact from stock-based awards. [Source: Tesla FY2025 Form 10-K, Note 2, p. 61]

## What must be proven

- Energy-storage growth and margin expansion must remain durable despite tariffs, product mix, and deployment timing. *Analyst judgment.*
- Automotive delivery growth, manufacturing utilization, and pricing must support automotive margins while Tesla funds new products and AI programs. *Analyst judgment.*
- Autonomy, robotics, and AI-related investments must earn attractive incremental returns rather than merely increase capital intensity. *Analyst judgment.*
- The appropriate enterprise-to-equity bridge treatment for lease liabilities, noncontrolling interests, and other claims remains **UNSOURCED** because the class convention was not provided.

## Research priorities

Review subsequent Tesla filings for vehicle deliveries, automotive gross margin, average selling price, regulatory-credit revenue, factory utilization, energy-storage unit economics, and capital expenditure. Assess the timing, regulatory feasibility, and economics of autonomy, Robotaxi, and other AI initiatives. *Analyst research plan.*

## Falsification question

**Partner-generated question — answer intentionally omitted:** What evidence would show that Tesla's expected future growth and profitability are not strong enough to justify its current market valuation?

## Investment call

**watch-defer.** Tesla's liquidity and energy-storage momentum are constructive, but declining automotive revenue and earnings, together with a planned step-up in AI-related capital spending, require more evidence before initiating. *Analyst judgment, informed by Tesla FY2025 Form 10-K, Item 7, pp. 31, 37, 41–42.*

This call would change to **initiate-buy** if Tesla demonstrates sustained improvement in automotive economics and evidence that energy and AI investments generate attractive returns on incremental capital. *Analyst judgment.*

## What changed from Edition A

| Item | Edition A said | Edition B / final model says | Why it changed | Source |
|---|---|---|---|---|
| Investment call | Watch / defer | **Watch** / **defer** (unchanged) | The model confirms the caution: every method values TSLA far below the share price | `Project 1/README.md` |
| Valuation | No valuation; numerical bridge left open | **$15–$20 per share** (BASE DCF $16.54); methods span $6.84–$39.10 vs. $356.09 (Sep 1, 2026) | Built a five-year FCFF DCF driven by a segment forecast, plus P/E and EV/EBITDA comparables | `valuation.py`, `comps.py` |
| FY2025 energy revenue | $12,270 million | **$12,771 million** | Edition A figure did not match the 27% growth it cited; corrected from the 10-K | FY2025 10-K, Item 7 |
| Debt in the bridge | $8,180 million principal | **$8,376 million** (debt + finance leases, balance sheet) | Use the balance-sheet amount, including finance leases; operating leases not treated as debt | FY2025 10-K balance sheet, p. 49 |
| Other bridge claims | Leases, NCI, other claims “require the class convention” | **Subtract NCI $670M and redeemable NCI $58M; add cash, short-term investments, and digital assets** | Resolved each open bridge line | FY2025 10-K balance sheet, p. 49 |
| Base year | Not normalized | **Normalized FY2025 operating income $4,849M** (reported $4,355M + $494M restructuring) | Restructuring is a one-time cost | FY2025 10-K income statement |
| Energy storage | Growth and margin must prove durable | **Energy margin 30% → 34%** in the base case; massive Megapack commercialization is a trigger toward initiate | Margin rose from 26.2% (FY2024) to 29.8% (FY2025) | FY2025 10-K, Item 7 |
| AI / Robotaxi | Returns on AI investment unproven | **Excluded from BASE; UPSIDE adds Robotaxi/FSD to $40B revenue →** | Shows how much the share price depends | `valuation.py` |

| Item | Edition A said | Edition B / final model says | Why it changed | Source |
|---|---|---|---|---|
| | | **$39.10/share**; the price implies about $602B of 2030 autonomy revenue | on autonomy | |
| Falsification question | Answer intentionally omitted | **Answered below** | | |

## Answer to the falsification question

**Question (partner-generated):** What evidence would show that Tesla's expected future growth and profitability are not strong enough to justify its current market valuation?

**Answer:** My valuation would be wrong if Tesla does not grow enough to reach the expectations in the reverse DCF. If Robotaxi revenue stays far below about $602 billion or Tesla cannot get close to $183 billion in FCFF by 2030, it would show that the current stock price is hard to support.